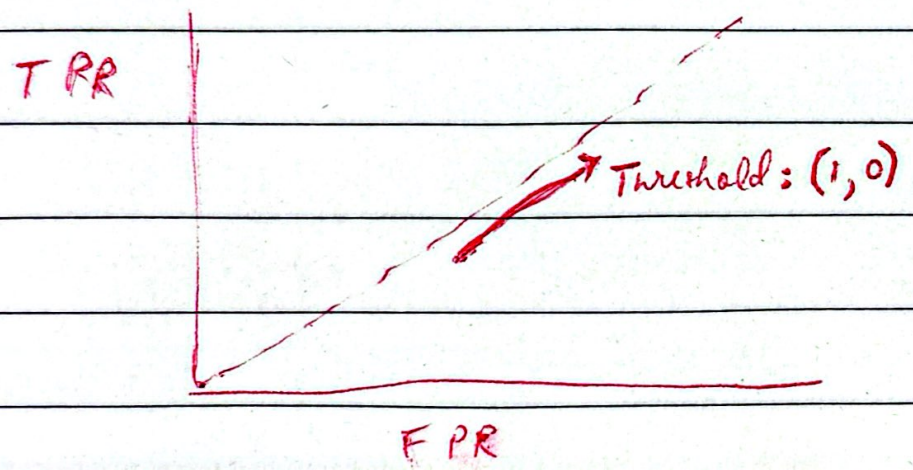
$$\text{Recall} = TPR = \frac{TP}{\underbrace{TP + FN}_{\text{total positives}}}$$

$$\text{Precision} = FPR = \frac{FP}{\underbrace{FP + TN}_{\text{Total negatives}}}$$

$$\text{F1-score} = \frac{2}{\text{Recall}^{-1} + \text{Precision}^{-1}} = \frac{2(\text{Rec})(\text{Prec})}{\text{Rec} + \text{Prec}}$$

AUC : TPR and FPR are all confidence Thresholds.



At high confidence threshold, you get low TPR & low FPR. At low confidence, you get high TPR & high FPR